

KONNERUD, Norway

WMO: 014770

Lat: 59.7128N

Lon: 10.1458E

Elev: 633

StdP: 14.36

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.5	3.4	-6.5	4.0	-0.8	-1.5	5.2	5.8	15.1	34.3	12.0	31.3	2.2	310	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	79.0	62.0	75.9	60.8	72.8	59.0	64.9	73.5	63.2	71.0	61.5	69.1	5.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	85.1	67.8	60.2	79.8	66.0	58.5	74.8	64.1	30.2	73.7	29.0	70.4	27.7	69.6	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.1	10.3	8.4	-4.4	83.2	5.3	3.5	-8.2	85.7	-11.3	87.8	-14.2	89.7	-18.0	92.3		
			-5.0	67.2	5.4	2.9	-8.9	69.3	-12.1	70.9	-15.1	72.5	-19.1	74.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	23.0	26.6	32.4	40.4	50.4	57.0	62.1	59.3	51.5	42.2	32.8	24.9
	DBStd	15.16	9.15	9.14	7.39	5.88	7.08	4.63	4.77	4.68	4.98	5.86	7.40	10.13
	HDD50	4001	836	656	546	292	83	2	0	0	41	249	517	779
	HDD65	8446	1301	1076	1011	737	452	243	116	187	404	708	967	1244
	CDD50	1075	0	0	0	5	96	214	377	289	87	7	0	0
	CDD65	43	0	0	0	0	1	4	27	11	0	0	0	0
	CDH74	473	0	0	0	0	36	62	272	102	1	0	0	0
	CDH80	55	0	0	0	0	3	7	38	7	0	0	0	0
Wind	WSAvg	3.4	3.1	3.4	3.7	3.8	3.8	3.9	3.6	3.2	3.2	3.2	3.0	3.2
Precipitation	PrecAvg	37.7	3.0	1.9	1.9	2.3	2.6	3.1	3.4	3.7	4.0	4.4	4.4	2.8
	PrecMax	57.1	6.8	5.7	5.3	6.4	5.4	7.3	6.8	6.7	12.4	10.9	18.2	5.9
	PrecMin	27.7	0.3	0.3	0.0	0.3	0.4	0.4	0.3	0.8	1.5	1.1	0.8	0.7
	PrecStd	6.0	1.8	1.1	1.4	1.5	1.3	1.6	1.5	1.4	2.3	2.4	3.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	49.0	58.9	66.0	78.4	80.1	83.8	80.8	71.3	60.9	52.6	48.3
		MCWB	39.0	40.9	47.0	52.4	59.2	62.0	64.6	60.8	59.7	53.2	47.1	43.7
	2%	DB	40.9	44.5	52.4	60.9	73.0	75.3	80.0	77.3	67.3	56.8	49.5	44.0
		MCWB	35.9	37.8	42.3	49.0	56.7	59.1	63.0	62.3	56.4	50.6	44.5	40.6
	5%	DB	37.9	41.4	47.2	56.0	68.7	71.5	76.9	73.6	63.8	54.4	45.8	40.7
		MCWB	34.1	35.8	38.8	45.2	54.3	56.8	61.5	60.4	54.4	49.7	42.8	37.1
	10%	DB	35.2	38.7	43.4	52.1	64.2	68.1	73.7	70.0	60.8	52.1	43.1	38.2
		MCWB	32.8	34.3	36.3	42.9	52.3	54.9	60.0	58.8	53.4	48.1	40.4	35.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.2	42.0	47.5	54.3	62.1	64.2	68.1	66.4	62.0	55.7	49.1	44.5
		MCDB	43.5	47.6	57.3	63.4	71.9	75.3	77.5	74.6	67.4	59.0	50.9	47.5
	2%	WB	37.4	39.2	43.0	50.0	59.2	61.4	65.8	64.3	58.7	53.0	45.5	40.8
		MCDB	40.5	43.0	51.2	59.5	68.4	70.5	75.7	71.2	64.1	55.0	47.9	42.9
	5%	WB	34.9	36.9	40.1	46.9	56.5	59.3	64.0	62.5	56.8	51.1	43.2	37.9
		MCDB	36.9	40.1	45.7	53.7	65.7	67.7	72.4	69.3	61.3	53.4	45.4	39.8
	10%	WB	32.9	35.1	37.4	44.2	53.9	57.3	62.3	60.6	55.2	49.0	40.9	35.3
		MCDB	34.7	37.8	42.6	50.4	61.7	64.8	69.9	67.8	59.2	51.7	42.9	37.4
Mean Daily Temperature Range	5% DB	MDBR	10.7	13.3	16.2	17.6	18.3	17.2	18.2	18.1	15.9	12.9	9.5	10.7
		MCDBR	13.7	16.4	22.2	25.5	26.4	23.9	24.2	22.9	22.3	17.1	14.7	13.4
		MCWBR	10.7	11.5	13.2	13.9	12.5	11.3	10.3	9.8	12.1	11.3	11.6	11.3
	5% WB	MCDBR	12.4	14.1	21.3	23.1	22.6	19.5	21.0	18.9	17.3	13.9	12.4	12.5
		MCWBR	10.4	10.8	13.6	13.6	11.8	11.3	10.5	10.4	11.1	11.0	11.1	11.1
Clear-Sky Solar Irradiance	taub	0.251	0.291	0.315	0.356	0.354	0.345	0.365	0.358	0.328	0.315	0.271	0.225	
	taud	2.229	2.316	2.357	2.355	2.440	2.484	2.445	2.458	2.533	2.492	2.316	2.135	
	Ebn at Noon	176	230	261	268	278	282	272	266	257	222	168	138	
	Edh at Noon	13	21	27	32	32	31	32	29	23	18	12	9	
All-Sky Solar Radiation	RadAvg	97	295	772	1227	1558	1777	1576	1243	823	371	129	65	
	RadStd	19	57	76	171	170	134	203	95	101	71	34	7	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (53 neighbors)	N/A	N/A	+2.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page